

Module 4

Introduction to Quantum Mechanics and Quantum Computing

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24 Nov 2025

- 1 Quantum Principles: Uncertainty & Superposition
- 2 Math Primer: Vectors & Linear Algebra
- 3 Mathematical Framework of Quantum Mechanics
- 4 Introduction to Quantum Computing

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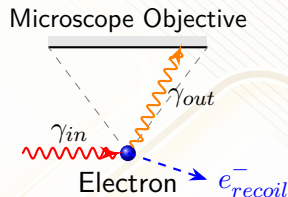
Heisenberg's Microscope Thought Experiment

Key insight: Measuring a quantum system inevitably disturbs it.

To “see” an electron, we need light (photons). The trade-off:

- ▶ High resolution $\rightarrow$ short wavelength ($\lambda \downarrow$)
- ▶ Short $\lambda \rightarrow$ high energy ($E \uparrow$)
- ▶ High energy photons $\rightarrow$ massive momentum kick
- ▶ **Result:** Precise position knowledge destroys momentum knowledge.

$$\Delta x \approx \frac{\lambda}{2 \sin \theta}, \quad \Delta p \approx \frac{h}{\lambda} \sin \theta$$



$$\Delta x \cdot \Delta p \geq \frac{\hbar}{2}$$

Where:

- ▶ Δx = uncertainty in position
- ▶ Δp = uncertainty in momentum
- ▶ $\hbar = \frac{h}{2\pi}$ = reduced Planck's constant

Physical meaning: It's fundamentally impossible to know both exact position and exact momentum simultaneously.

Heuristic (microscope) estimate:

$$\Delta x \approx \frac{\lambda}{2 \sin \theta}, \quad \Delta p \approx \frac{h}{\lambda} \sin \theta \quad \Rightarrow \quad \Delta x \Delta p \approx \frac{h}{2} \sim \frac{\hbar}{2\pi}.$$

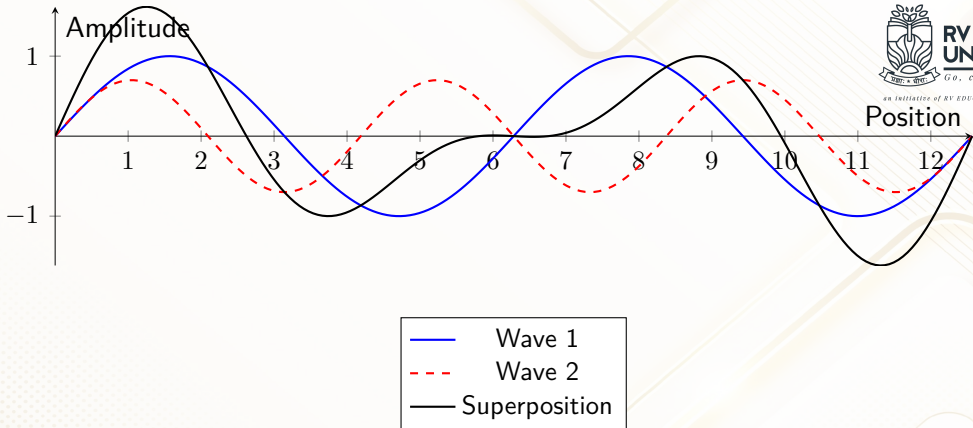
A quantum system can exist in multiple states **simultaneously**:

$$|\psi\rangle = a|0\rangle + b|1\rangle$$

Classical bit: Either 0 OR 1.

Quantum bit (qubit): 0 AND 1 *simultaneously* (with amplitudes a, b).

Wave Interference: Constructive vs Destructive



Constructive interference: Amplitudes add → higher probability.

Destructive interference: Amplitudes cancel → lower probability.

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Why We Need Mathematics for Quantum Mechanics



- ▶ **Vectors** represent quantum states
- ▶ **Vector length** represents probability (must sum to 1)
- ▶ **Dot product** measures similarity between states
- ▶ **Matrices** represent quantum operations (gates)
- ▶ **Complex numbers** essential for quantum amplitudes

Real-world Connection

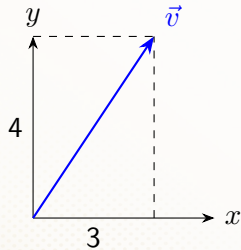
Just like classical computers use binary (0/1), quantum computers use vectors to represent qubit states. The mathematics ensures probabilities make sense and operations are physically possible.

A vector has magnitude and direction:

$$\vec{v} = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$$

Magnitude (length):

$$|\vec{v}| = \sqrt{3^2 + 4^2} = 5$$



Everyday Example

Think of a vector as a displacement: "walk 3 meters east, 4 meters north" — you end up 5 meters from start.

Quantum State Vectors & Complex Numbers

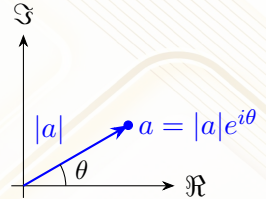
Quantum states are vectors with **complex** components:

$$|\psi\rangle = \begin{bmatrix} a \\ b \end{bmatrix} = a|0\rangle + b|1\rangle, \quad a, b \in \mathbb{C}.$$



Why Complex Numbers?

- ▶ **Magnitude** (aa^*) determines probability.
- ▶ **Phase** ($e^{i\theta}$) determines **interference**.
- ▶ Evolution is *unitary* (rotation), preserving total probability.



Phase θ rotates the vector but keeps length (prob) constant.

Phase Matters!

$|+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$ and $|-\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$ behave differently under interference.

Common states:

$$|0\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad |1\rangle = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \quad |+\rangle = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \quad |-\rangle = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$



Example with Complex Phase

$$|\psi\rangle = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ i \end{bmatrix} \Rightarrow P(0) = P(1) = \frac{1}{2}$$

Same probabilities as $|+\rangle$, but different phase affects interference.

Phase Visualization

The phase $e^{i\theta}$ can be visualized as a rotation in the complex plane. States with different phases will interfere constructively or destructively when combined.

Normalization: Total Probability Must Be 1

A quantum state

$$|\psi\rangle = \begin{bmatrix} a \\ b \end{bmatrix}, \quad a, b \in \mathbb{C}$$

must satisfy the **normalization condition**:

$$|a|^2 + |b|^2 = 1.$$



Why This Condition?

- ▶ $|a|^2$ = probability of measuring $|0\rangle$
- ▶ $|b|^2$ = probability of measuring $|1\rangle$
- ▶ Total probability of all outcomes must be **1**

Renormalizing an Unnormalized State

$$\text{If } |\chi\rangle = \begin{bmatrix} 2 \\ 1 \end{bmatrix} \text{ then } \|\chi\| = \sqrt{5} \text{ and } |\chi_{\text{norm}}\rangle = \frac{1}{\sqrt{5}} \begin{bmatrix} 2 \\ 1 \end{bmatrix}.$$

Properly Normalized States

$$|\psi\rangle = \frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle \Rightarrow P(0) = P(1) = \frac{1}{2}$$

$$|\phi\rangle = \frac{1}{2}|0\rangle + \frac{\sqrt{3}}{2}|1\rangle \Rightarrow \frac{1}{4} + \frac{3}{4} = 1$$

States That Need Normalization

$$|0\rangle + |1\rangle, \quad 2|0\rangle + 2|1\rangle, \quad 3|0\rangle + 4i|1\rangle$$

Inner Product: Measuring Quantum Similarity



What is the Inner Product?

- ▶ **Measures similarity** between quantum states
- ▶ **Complex number** with magnitude and phase
- ▶ **Key formula:** $|\langle\phi|\psi\rangle|^2 = \text{probability}$

Key Intuitions

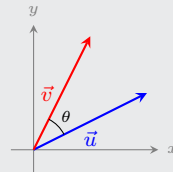
- ▶ **Same state:** $\langle\psi|\psi\rangle = 1$
- ▶ **Orthogonal:** $\langle\phi|\psi\rangle = 0$
- ▶ **Partial overlap:** $0 < |\langle\phi|\psi\rangle|^2 < 1$

Classical Dot Product (Warm-up)

$$\vec{u} \cdot \vec{v} = u_1v_1 + u_2v_2$$

- ▶ Same direction: large positive
- ▶ Opposite direction: large negative
- ▶ Perpendicular: zero (orthogonal)

Classical Vector Alignment



$$\vec{u} \cdot \vec{v} = \|\vec{u}\| \|\vec{v}\| \cos \theta$$



Quantum Inner Product

For $|\phi\rangle = \begin{pmatrix} a \\ b \end{pmatrix}$ and $|\psi\rangle = \begin{pmatrix} c \\ d \end{pmatrix}$:

$$\langle\phi|\psi\rangle = a^*c + b^*d$$

- ▶ Complex conjugation (a^*) accounts for quantum phases
- ▶ Magnitude squared $\Rightarrow$ measurement probability

Concrete Examples

- ▶ $\langle 0|0\rangle = 1$
- ▶ $\langle 0|1\rangle = 0$
- ▶ $\langle ++\rangle = 1$
- ▶ $\langle +|- \rangle = 0$

Physical Interpretation

$|\langle\phi|\psi\rangle|^2 =$ probability that measuring $|\psi\rangle$ gives result corresponding to $|\phi\rangle$.

Orthogonal States = Distinguishable Outcomes

Two states are orthogonal if:

$$\langle \phi | \psi \rangle = 0.$$

Example: $|0\rangle$ and $|1\rangle$ are orthogonal:

$$\langle 0 | 1 \rangle = \begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = 0$$

Physical meaning: Orthogonal states can be perfectly distinguished by measurement.

More Orthogonal Pairs

- ▶ $|+\rangle$ and $|-\rangle$: $\langle + | - \rangle = 0$
- ▶ Any computational basis states are orthogonal

Basis Vectors: Building Blocks of Quantum States

A basis is a set of vectors that can represent any state:

Computational basis:

$$|0\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad |1\rangle = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

Any qubit state:

$$|\psi\rangle = a|0\rangle + b|1\rangle, \quad a = \langle 0|\psi\rangle, \quad b = \langle 1|\psi\rangle.$$



Different Bases

- ▶ **Computational basis:** $\{|0\rangle, |1\rangle\}$
- ▶ **Hadamard basis:** $\{|+\rangle, |-\rangle\}$

State: $|\psi\rangle = \frac{1}{\sqrt{3}}|0\rangle + \sqrt{\frac{2}{3}}|1\rangle$

Check normalization:

$$\left| \frac{1}{\sqrt{3}} \right|^2 + \left| \sqrt{\frac{2}{3}} \right|^2 = \frac{1}{3} + \frac{2}{3} = 1$$

Probabilities:

$$P(0) = \frac{1}{3}, \quad P(1) = \frac{2}{3}$$

Matrices transform quantum states:

Example - X gate (quantum NOT):

$$X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \quad X|0\rangle = |1\rangle.$$

Common Quantum Gates

X gate (NOT):

$$X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

Z gate (phase flip):

$$Z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

Hadamard gate:

$$H = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$

Y gate:

$$Y = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}$$

Quantum operations must preserve total probability. All valid quantum gates are represented by **unitary matrices**.

$$U^\dagger U = I.$$

Why Unitary?

- ▶ Preserve norm: $\|U|\psi\rangle\|^2 = \||\psi\rangle\|^2$
- ▶ Preserve inner products and phases
- ▶ Geometrically: rotations in complex space

Verifying Unitarity

For $X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$:

$$X^\dagger X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I$$

For $H = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$:

$$H^\dagger H = \frac{1}{2} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} = I$$

Quick Check: Normalization

Which of these states are normalized?

1 $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$

2 $\begin{bmatrix} \frac{1}{2} \\ \frac{\sqrt{3}}{2} \end{bmatrix}$

3 $\begin{bmatrix} 2 \\ 2 \end{bmatrix}$

4 $\begin{bmatrix} \frac{1+i}{2} \\ \frac{1-i}{2} \end{bmatrix}$

5 $\begin{bmatrix} 0.6 \\ 0.8 \end{bmatrix}$

Which of these states are normalized?

- 1 $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$
- 2 $\begin{bmatrix} \frac{1}{2} \\ \frac{\sqrt{3}}{2} \end{bmatrix}$
- 3 $\begin{bmatrix} 2 \\ 2 \end{bmatrix}$
- 4 $\begin{bmatrix} \frac{1+i}{2} \\ \frac{1-i}{2} \end{bmatrix}$
- 5 $\begin{bmatrix} 0.6 \\ 0.8 \end{bmatrix}$

Answers:

- 1 $1^2 + 1^2 = 2 \rightarrow$ No
- 2 $\frac{1}{4} + \frac{3}{4} = 1 \rightarrow$ Yes
- 3 $4 + 4 = 8 \rightarrow$ No
- 4 $\left|\frac{1+i}{2}\right|^2 + \left|\frac{1-i}{2}\right|^2 = \frac{2}{4} + \frac{2}{4} = 1 \rightarrow$ Yes
- 5 $0.6^2 + 0.8^2 = 1 \rightarrow$ Yes

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Before measurement: System in superposition

$$|\psi\rangle = a|0\rangle + b|1\rangle$$

During measurement: System "chooses" an outcome

$$P(0) = |a|^2, \quad P(1) = |b|^2.$$

After measurement: System collapses to definite state:

$$|\psi\rangle \rightarrow |0\rangle \quad \text{or} \quad |\psi\rangle \rightarrow |1\rangle.$$

$$|\psi\rangle = \sum_i a_i |i\rangle.$$

Measurement in basis $\{|i\rangle\}$:

$$P(i) = |\langle i|\psi\rangle|^2.$$

After measurement:

$$|\psi\rangle \longrightarrow |i\rangle \quad \text{with probability } P(i).$$

Key Features

- ▶ Probabilistic outcomes
- ▶ Measurement changes the state
- ▶ Only orthonormal basis states can be outcomes

Projective Measurement (Projection Operators)

A measurement is described by projectors

$$P_i = |i\rangle\langle i|, \quad P_i^\dagger = P_i, \quad P_i^2 = P_i, \quad \sum_i P_i = I.$$



Outcome probability:

$$P(i) = \langle \psi | P_i | \psi \rangle.$$

Post-measurement state:

$$|\psi\rangle \longrightarrow \frac{P_i |\psi\rangle}{\sqrt{\langle \psi | P_i | \psi \rangle}}.$$

Example

$P_0 = |0\rangle\langle 0|$, $P_1 = |1\rangle\langle 1|$. If $|\psi\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$ then $P(0) = P(1) = \frac{1}{2}$.

Observables: Operators That Correspond to Measurable Quantities

An observable is a Hermitian operator A with

$$A = A^\dagger.$$

Spectral decomposition:

$$A = \sum_i a_i |a_i\rangle\langle a_i|.$$

Probability of outcome a_i :

$$P(a_i) = |\langle a_i | \psi \rangle|^2.$$



Z observable

$$Z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} = +1 |0\rangle\langle 0| - 1 |1\rangle\langle 1|.$$

Hermitian Operators: Real Eigenvalues, Physical Measurements

A Hermitian $\Rightarrow A^\dagger = A$.

- ▶ Eigenvalues are real.
- ▶ Eigenvectors form an orthonormal basis (if non-degenerate).
- ▶ Only Hermitian operators represent observables.

Examples

$$X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}, \quad Z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

Pauli matrices X, Z are Hermitian and therefore valid observables.

Given observable A and state $|\psi\rangle$,

$$\langle A \rangle = \langle \psi | A | \psi \rangle.$$

Meaning: average of many repeated measurements on identical preparations.

Example

$$|\psi\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle), \quad Z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}.$$

$$\langle Z \rangle = \frac{1}{2}(1) + \frac{1}{2}(-1) = 0.$$

The Schrödinger equation:

$$i\hbar \frac{\partial}{\partial t} |\psi(t)\rangle = H |\psi(t)\rangle,$$

where H is the Hamiltonian.

Time-independent eigenproblem:

$$H|\psi\rangle = E|\psi\rangle, \quad |\psi(t)\rangle = |\psi(0)\rangle e^{-iEt/\hbar}.$$

Keypoint

Probability distribution doesn't change with time:

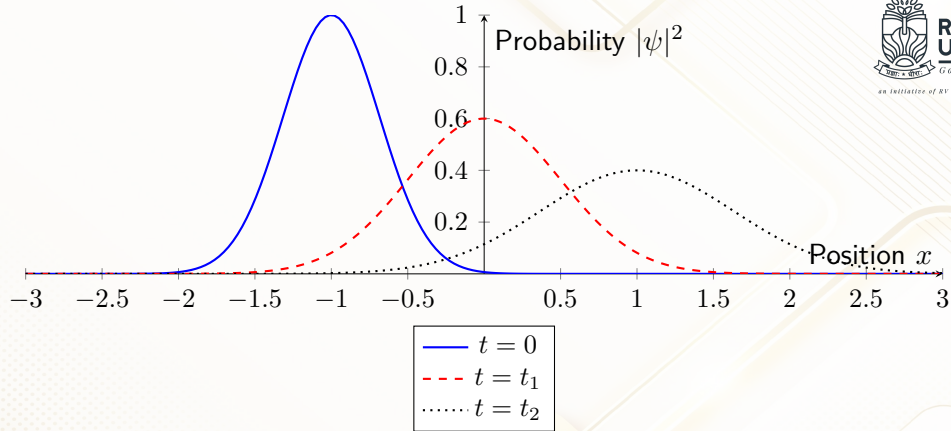
$$|\psi(x, t)|^2 = |\psi(x, 0)|^2$$

For $V(x) = 0$:

$$-\frac{\hbar^2}{2m} \frac{d^2\psi}{dx^2} = E\psi$$

Solutions: plane waves $\psi(x) = Ae^{ikx} + Be^{-ikx}$ with $k = \sqrt{2mE}/\hbar$. Note: plane waves are not normalizable over infinite domain.

Wave Packets: Localized Particles



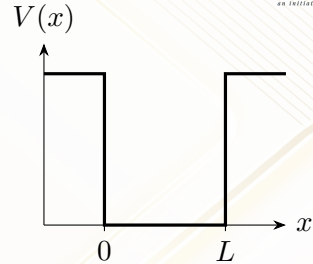
Wave packets: Superposition of multiple wavelengths create localized particles that spread over time.

Particle in a Box: Quantized Energy

Infinite potential well:

$$V(x) = \begin{cases} 0 & 0 < x < L \\ \infty & \text{otherwise} \end{cases}$$

Boundary: $\psi(0) = \psi(L) = 0$. Energy quantized.



Wavefunctions:

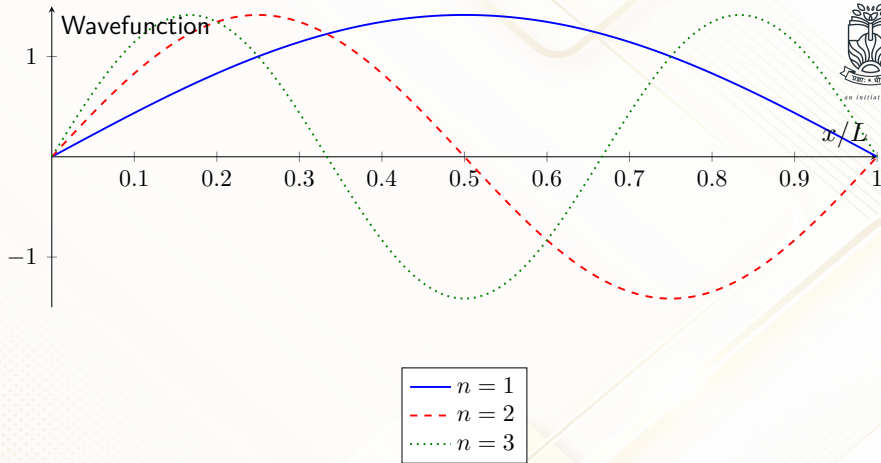
$$\psi_n(x) = \sqrt{\frac{2}{L}} \sin\left(\frac{n\pi x}{L}\right), \quad n = 1, 2, \dots$$

Energies:

$$E_n = \frac{n^2 \pi^2 \hbar^2}{2mL^2}.$$

Ground state has non-zero energy; nodes increase with n .

Wavefunctions for Particle in a Box



Each energy level has a distinct wavefunction with increasing number of nodes.

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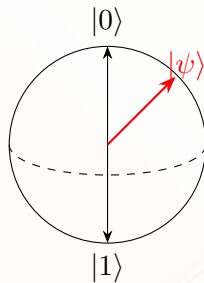
Qubit: The Quantum Bit

Classical bit:

- ▶ Values: 0 or 1
- ▶ State: discrete

Quantum bit (qubit):

- ▶ State: $|\psi\rangle = a|0\rangle + b|1\rangle$
- ▶ $a, b \in \mathbb{C}$, $|a|^2 + |b|^2 = 1$
- ▶ Continuous superposition; manipulated by unitary gates



Pauli-X: $X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$, $X|0\rangle = |1\rangle$.

Pauli-Z: $Z = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$, $Z|+\rangle = |-\rangle$.

Hadamard: $H = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$, $H|0\rangle = |+\rangle$.

All are unitary: $U^\dagger U = I$.

Hadamard Gate: Creating Superposition

$$H = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$

$$H|0\rangle = |+\rangle, \quad H|1\rangle = |-\rangle, \quad H^2 = I.$$

Two-qubit state:

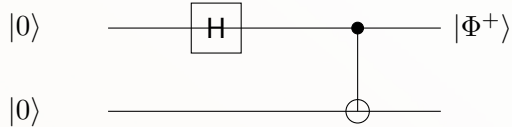
$$|\psi\rangle = a|00\rangle + b|01\rangle + c|10\rangle + d|11\rangle.$$

Bell state:

$$|\Phi^+\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}}.$$

Entanglement is correlation stronger than classical correlation.

Creating Entanglement: CNOT Gate



$$\text{Circuit: } |0\rangle|0\rangle \xrightarrow{H \otimes I} |+\rangle|0\rangle \xrightarrow{\text{CNOT}} \frac{|00\rangle + |11\rangle}{\sqrt{2}}.$$

Classical

Bits: 0 or 1
Deterministic ops
Cloning allowed
Local correlations

Quantum

Qubits: superpositions
Unitary (reversible) ops
No-cloning theorem
Entanglement

Summary: Key Quantum Concepts

- ▶ State representation: Vectors in Hilbert space
- ▶ Superposition and interference
- ▶ Measurement and collapse (projectors)
- ▶ Observables: Hermitian operators
- ▶ Schrödinger equation and time evolution
- ▶ Quantum gates and entanglement

- ▶ Griffiths, *Introduction to Quantum Mechanics*
- ▶ Feynman, *QED: The Strange Theory of Light and Matter*
- ▶ Nielsen & Chuang, *Quantum Computation and Quantum Information*
- ▶ IBM Qiskit tutorials, Microsoft QDK, Google Cirq